# AHMED MUBARAK

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## SUMMARY

**Automation and Control Systems Engineer (EIT)** with experience in **industrial automation, electrical control system design, PLC programming, and SCADA/HMI systems** across energy, water/wastewater, and critical infrastructure projects. Currently working at **AECOM as an I&C Automation Engineer**, developing **electrical schematics, panel layouts, I/O documentation, and automation system integrations** for large-scale engineering projects. Experienced in supporting **FAT, SAT, commissioning, and system startup**, with a strong ability to translate **P&IDs and machine requirements into reliable control system designs.**

## EDUCATION

**Bachelor of Automation Engineering Technology**, ***McMaster University*** ***Sept 2021 - Dec 2025***

- Advanced Diploma, Chemical Engineering Process Design**,** ***Mohawk College*** ***Sept 2021 - Dec 2023***
- Business Management Professional Certificate, ***McMaster University*** ***Accredited By (ACBSP)***

## RELEVANT WORK EXPERIENCE

**Junior Electrical Engineer (I&C) - EIT, AECOM** ***Nov 2025 - Present***

- Support electrical and I&C design for large projects by reviewing **P&IDs** and **translating process** requirements into **electrical drawing**s, including **control schematics**, **panel layouts**, and **IO lists.**
- Participate in **(FAT/SAT) testing**, **commissioning activities**, collaborating with multidisciplinary teams to ensure safe, reliable, and standards-compliant project delivery.
- Contribute to the design, review and development of **SCADA/HMI** systems and **PLC programming**, supporting process monitoring and control for municipal water and critical infrastructure facilities.
- Assist in the development of **electrical drawing packages** for construction, supporting **BOM** preparation**,** vendor package review, and coordination with multidisciplinary engineering teams.

**Automation & Controls Engineer (Co-op), JNE Group** ***May 2025 – Sept 2025***

- Designed, programmed, and commissioned **Allen-Bradley (Studio 5000)** and **Siemens (TIA Portal)** control systems for Dofasco, STELCO, Algoma Steel, and US Steel.
- Engineered the complete control system design for **US Steel**'s HDGL wastewater treatment plant, producing **AutoCAD schematics**, **panel layouts**, and a **+368-point I/O** list from **P&ID** analysis. Conducted **FAT**s for **control panels** and supported on-site **commissioning.**

**Control Systems Engineer, Midwest Services Canada** ***Nov 2024 – May 2025***

- Designed and implemented PLC (**Siemens**) based automation solutions for commercial buildings.
- Provided full engineering from design to implementation for the **Delta Marriott Hotel's networked compressor control system,** including **electrical design**, **panel building**, and **on-site commissioning.**

**Electrical/Controls Field Engineer (Co-op), GE Vernova (General Electric)** ***Jan 2024 - Sept 2024***

- Supported commissioning, maintenance, and upgrades of control (**DCS**) systems for gas turbines, steam turbines, and generators during ***Ontario****'s 2024 spring* outage season across five **power stations**.
- Strongly participated in internal and external planning meetings, working alongside senior project managers and Canada's lead engineers to support the execution and scheduling of projects.

**Electric Design Engineer (Co-op), Salford Group (Linamar Corporation)** ***May 2023 - Sept 2023***

- Designed electrical harnesses, routing diagrams, and control panel schematics/layouts in **AutoCAD Electrical** for agricultural vehicles, improving manufacturability and design accuracy.

## SKILLS

**Instrumentation & Control Systems:** Allen-Bradley PLCs (Studio 5000), Siemens PLCs (TIA Portal, LOGO!), SCADA (GE [Proficy, Cimplicity, iFIX], Siemens WinCC, FactoryTalk, Ignition, AVEVA), HMI development
**Engineering & Design:** AutoCAD, EPLAN, Revit, electrical design packages, P&IDs, I/O lists, Loop Diagram
**Industrial Expertise:** Industrial communication protocols (EtherNet/IP, PROFINET, Modbus TCP/IP, BACnet, HART), instrumentation sizing, FAT/SAT, commissioning, troubleshooting, CSA & ISA standards

## ACADEMIC EXPERIENCE

**Automation LAB Research Assistant (Part-Time) McMaster University** ***Sept 2024 – May 2025***

- Designing and building an RC car platform with advanced control systems.
- Programmed and wired a conveyor line using **Allen-Bradley PLC** and **HMI** using **Factory Talk View**.
- Configured and tuned (**VFDs**) for precise motor control in a conveyor system.

**Allen Bradley PLC Advanced (3PL3), TA, McMaster University** ***Sept 2025 - Dec 2025***
**Allen Bradley PLC (2PL3), TA, McMaster University** ***Jan 2025 - April 2025***
**Chemical Engineering Process Design (2EC3), TA, McMaster University** ***Jan 2025 - April 2025***
**Math 1 (1MC3), TA, McMaster University** ***Sept 2024 - Dec 2024***

## PROJECTS, EXTRACURRICULARS & CLUBS

**HVAC Air Compressors Control System – Delta Ottawa Hotel (Capstone)** ***Jan 2025 – March 2025***

- Led the end-to-end design and implementation of a dual air compressor building automation system powered by a **Siemens LOGO! PLC** and **CSA-compliant custom panels**.
- Designed full system architecture, Electrical schematics, power distribution, control circuits, and protection schemes for fabrication and **ESA/ESAFE** certification.
- Programmed control logic for automatic/manual operation, load balancing, runtime tracking, and maintenance scheduling, extending compressor lifecycle and reliability.
- Integrated advanced alarm handling (leak detection, low pressure, abnormal operation) and automated daily engineer reports, providing operators with real-time visibility of system health.
- Managed project planning, supplier coordination, commissioning, and engineer training, delivering a fully certified, standards-compliant solution that reduced downtime and improved operational safety.

**SMC IPC-202A Bottling Station – Industrial Process Control Training** ***Sept 2025 – Dec 2025***

- Working in a 3-person team to install, wire, and program a bottling station with pneumatic actuators, **Omron PLC**, and **HM**I.
- Mapping and testing all **I/O** devices, integrating safety interlocks, and verifying actuator response for accurate filling and capping cycles.
- Programming automatic and manual modes in **Sysmac Studio**, implementing alarms, and developing HMI screens for operator monitoring and control.

**HVAC Control System with Schneider M172** ***(Learning Project)*** ***Sept 2025 – Dec 2025***

- Developing a small-scale HVAC automation system using the Schneider M172 programmable controller to replicate real building automation functions.
- Designing and wiring a professional-grade control panel including I/O modules, an Ethernet switch, and a motorized water valve actuator.
- Programming sequences of operation for heating/cooling loops, alarms, and monitoring, while documenting logic flow, panel diagrams, and system integration.

**Automated Swimming Pool System – Capstone Project 1** ***Jan 2025 – April 2025***

- Developed an **Allen-Bradley PLC**-based **control system** to fully automate water level, temperature regulation, chemical dosing, and pH balancing, ensuring optimal swimming pool conditions.
- Designed and built an **industrial-standard control panel**, incorporating safety interlocks, relay logic, and precise wiring to ensure system reliability and compliance with industry standards.
- Created a **FactoryTalk View HMI** for real-time monitoring, user-friendly operator control, and seamless system integration, allowing for remote adjustments and troubleshooting.

**Battery Workforce Challenge (Club)** ***May 2024 – Dec 2024***

- Contributed to the BMS Hardware team by designing and integrating advanced battery systems, enhancing functionality and efficiency, and fostering innovative problem-solving.
- Performed battery performance testing, focusing on hardware integration and cell characterization.